Bruno Cerda Mardini

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TECHNICAL SKILLS

Programming Languages Python, C, C++
Tools PyTorch, Gymnasium, OpenMP, MPI, Git

RESEARCH EXPERIENCE

Research Assistant, *Vision and Language Group*, CENIA (Supervised by Professor *Hans Lobel*) 2024 – Present

- Engineered a decoder-free, sample-efficient world model using the Sketched Isotropic Gaussian Regularizer to prevent representation collapse, achieving 2x speedup over the STORM baseline on Atari-100k (**ICLR 2027, under review**).
- Reimplemented from scratch a SOTA world model called STORM in PyTorch (GitHub), creating the training and evaluation pipelines that were leveraged as the foundational backbone for our own new decoder-free world model.

Research Assistant, *Neuro-Symbolic AI Group*, CENIA (Supervised by Professor *Cristóbal Rojas*) 2022 – 2024

- Designed and developed a novel algorithm to quantify the complexity of three-dimensional data from mice retinal time series, employing HPC with multithreading to handle intensive computational demands.
- Contributed high-performance algorithms to co-author the preprint "*XBP1s as a Therapeutic Target to Preserve Retinal Function During Aging and Neurodegeneration*" (bioRxiv).

EDUCATION

Master's Degree in Machine Learning, Pontifical Catholic University of Chile 2025 – Present (expected 2027-1)

- Focused on World Models and Reinforcement Learning

Bachelor's Degree in Engineering, Pontifical Catholic University of Chile 2021 – 2025

- Major in Data Science

LANGUAGES

Spanish *Native*
English *Professional Working Proficiency*